

On the Implicit Function Theorem and Consequences: Part II

Dr Francois Ebobisse

October 2024

1 Dini's theorem for systems of equations

Let us consider the following system of two equations with fourth unknowns x, y, s, t .

$$\begin{cases} F(x, y, s, t) = 0 \\ G(x, y, s, t) = 0 \end{cases} \quad (1)$$

Now assume that the first equation defines implicitly s as a function of x, y, t . That is,

$$s = h(x, y, t) \quad (2)$$

and let us substitute this expression of s in the second equation and get the following equation

$$H(x, y, t) := G(x, y, h(x, y, t), t) = 0. \quad (3)$$

Now let us assume that (3) defines implicitly t as a function of the variables x and y . That is,

$$t = g(x, y). \quad (4)$$

We substitute this expression of t in (2) and get s as a function of x and y . That is,

$$s = f(x, y) = h(x, y, g(x, y)). \quad (5)$$

So, from the system (1) we formally obtained the variables s and t as function of x and y . We then say that the functions $s = f(x, y)$ and $t = g(x, y)$ are defined implicitly from the system (1). Therefore the system (1) takes the form

$$\begin{cases} F(x, y, f(x, y), g(x, y)) = 0 \\ G(x, y, f(x, y), g(x, y)) = 0 \end{cases} \quad (6)$$

Now let us assume that we can partially differentiate the equations in (6) with respect to x . So we obtain the following system of linear equations with unknowns f_x and g_x

$$\begin{cases} F_x + F_s f_x + F_t g_x = 0 \\ G_x + G_s f_x + G_t g_x = 0 \end{cases} \quad (7)$$

If we assume that the determinant

$$\begin{vmatrix} F_s & F_t \\ G_s & G_t \end{vmatrix} = F_s G_t - F_t G_s \neq 0,$$

then we get

$$f_x = - \frac{\begin{vmatrix} F_x & F_t \\ G_x & G_t \end{vmatrix}}{\begin{vmatrix} F_s & F_t \\ G_s & G_t \end{vmatrix}} \quad \text{and} \quad g_x = - \frac{\begin{vmatrix} F_s & F_x \\ G_s & G_x \end{vmatrix}}{\begin{vmatrix} F_s & F_t \\ G_s & G_t \end{vmatrix}}. \quad (8)$$

Analogously we obtain

$$f_y = - \frac{\begin{vmatrix} F_y & F_t \\ G_y & G_t \end{vmatrix}}{\begin{vmatrix} F_s & F_t \\ G_s & G_t \end{vmatrix}} \quad \text{and} \quad g_y = - \frac{\begin{vmatrix} F_s & F_y \\ G_s & G_y \end{vmatrix}}{\begin{vmatrix} F_s & F_t \\ G_s & G_t \end{vmatrix}}. \quad (9)$$

Therefore, the condition we will need in order to get the variables s and t as functions of x and y from the system (1) around a given point (x_0, y_0, s_0, t_0) is that

$$\begin{aligned} \det \left(\frac{\partial(F, G)}{\partial(s, t)}(x_0, y_0, s_0, t_0) \right) &= \begin{vmatrix} F_s(x_0, y_0, s_0, t_0) & F_t(x_0, y_0, s_0, t_0) \\ G_s(x_0, y_0, s_0, t_0) & G_t(x_0, y_0, s_0, t_0) \end{vmatrix} \\ &= F_s(x_0, y_0, s_0, t_0)G_t(x_0, y_0, s_0, t_0) - F_t(x_0, y_0, s_0, t_0)G_s(x_0, y_0, s_0, t_0) \neq 0. \end{aligned}$$

Notations: Let $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_m)$. We denote

$$(x, y) = (x_1, \dots, x_n, y_1, \dots, y_m) \in \mathbb{R}^{n+m}.$$

Now let us consider the system of equations

$$\begin{cases} F_1(x, y) = 0 \\ F_2(x, y) = 0 \\ \dots \\ F_m(x, y) = 0 \end{cases} \quad (10)$$

We have the following theorem.

Theorem 1.1 *Let Ω be an open subset of $\mathbb{R}^{n+m}$ and $F = (F_1, \dots, F_m) : \Omega \rightarrow \mathbb{R}^m$ be a function of class $\mathcal{C}^1$. Let (a, b) with $a = (a_1, \dots, a_n)$ and $b = (b_1, \dots, b_m)$ be a point in Ω such that*

$$F(a, b) = 0 \quad \text{and} \quad \det \left(\frac{\partial(F_1, \dots, F_m)}{\partial(y_1, y_2, \dots, y_m)}(a, b) \right) \neq 0. \quad (11)$$

Then there exist $\delta, \beta > 0$ and a unique function $f = (f_1, \dots, f_m) : B(a, \delta) \rightarrow B(b, \beta)$ such that $f(a) = b$ and

$$F(x, f(x)) = 0 \quad \forall x \in B(a, \delta). \quad (12)$$

Moreover, f is a function of class $\mathcal{C}^1$ and, for any $x \in B(a, \delta)$, the jacobian matrix of f at x is given by

$$\frac{\partial(f_1, \dots, f_m)}{\partial(x_1, x_2, \dots, x_n)}(x) = - \left[\frac{\partial(F_1, \dots, F_m)}{\partial(y_1, y_2, \dots, y_m)}(x, f(x)) \right]^{-1} \frac{\partial(F_1, \dots, F_m)}{\partial(x_1, x_2, \dots, x_n)}(x, f(x)). \quad (13)$$

The proof is omitted!

Exercise 1.2 Consider the functions

$$F(x, y, z) := -e^x - 2y + \arctan z + \sin z + 1 \quad \text{and} \quad G(x, y, z) := 2x + \cos y - e^z \quad \forall (x, y, z) \in \mathbb{R}^3.$$

(a) Show that the system of equations

$$\begin{cases} F(x, y, z) = 0 \\ G(x, y, z) = 0 \end{cases}$$

defines around the point $(0, 0, 0)$ a unique regular curve $\mathcal{C}$ with parametric equations

$$\begin{cases} y = f(x), \\ z = g(x). \end{cases}$$

(b) Write down the parametric equations of the tangent line to the curve $\mathcal{C}$ at the point $(0, 0)$.

Solutions:

(a) Notice that the function F and G are of class $\mathcal{C}^\infty$. Also

$$\begin{cases} F(0, 0, 0) = 0 \\ G(0, 0, 0) = 0 \end{cases} \quad \text{and} \quad \frac{\partial(F, G)}{\partial(y, z)} = \begin{vmatrix} F_y(x, y, z) & F_z(x, y, z) \\ G_y(x, y, z) & G_z(x, y, z) \end{vmatrix}.$$

$$F_y = -2, \quad F_z = \frac{1}{1+z^2} + \cos z, \quad G_y = -\sin y, \quad G_z = -e^z.$$

Hence,

$$\frac{\partial(F, G)}{\partial(y, z)}(0, 0, 0) = \begin{vmatrix} F_y(0, 0, 0) & F_z(0, 0, 0) \\ G_y(0, 0, 0) & G_z(0, 0, 0) \end{vmatrix} = \begin{vmatrix} -2 & 2 \\ 0 & -1 \end{vmatrix} = 2 \neq 0.$$

Hence, by the implicit function theorem for system of equations (Theorem 1.1), there exist $\delta > 0$ and $\beta > 0$ and a unique vector function

$$(f, g) : (-\delta, \delta) \rightarrow B((0, 0), \beta)$$

of class $\mathcal{C}^\infty$ such that

$$\forall x \in (-\delta, \delta), \quad \forall (y, z) \in B((0, 0), \beta), \quad \begin{cases} F(x, y, z) = 0 \\ G(x, y, z) = 0 \end{cases} \Leftrightarrow \begin{cases} y = f(x) \\ z = g(x) \end{cases}$$

Therefore, we obtain the curve $\mathcal{C}$ with parametric equations

$$y = f(x), \quad z = g(x) \quad \text{with} \quad x \in (-\delta, \delta)$$

or described by the vector function $\vec{r} : (-\delta, \delta) \rightarrow \mathbb{R}^3$

$$\vec{r}(x) := (x, f(x), g(x)) \quad \forall x \in (-\delta, \delta).$$

- (b) From 2AC, the tangent line to the curve $\mathcal{C}$ at the point $(0, 0, 0)$ has direction $\vec{r}'(0)$. We find that

$$\vec{r}'(x) = (1, f'(x), g'(x)).$$

We know from Theorem 1.1 that

$$\begin{pmatrix} f'(x) \\ g'(x) \end{pmatrix} = - \left[\frac{\partial(F, G)}{\partial(y, z)}(x, f(x), g(x)) \right]^{-1} \begin{pmatrix} F_x(x, f(x), g(x)) \\ G_x(x, f(x), g(x)) \end{pmatrix}$$

We find that

$$\begin{aligned} F_x = -e^x, \quad G_x = 2 \quad \Rightarrow \quad \begin{pmatrix} f'(0) \\ g'(0) \end{pmatrix} &= \begin{pmatrix} -2 & 2 \\ 0 & -1 \end{pmatrix}^{-1} \begin{pmatrix} -1 \\ 2 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} -1 & -2 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} -3/2 \\ -2 \end{pmatrix}. \end{aligned}$$

So, the tangent line to the curve $\mathcal{C}$ at the point $(0, 0, 0)$ has direction

$$\vec{r}'(0) = (1, -3/2, -2).$$

Therefore, the tangent line to the curve $\mathcal{C}$ at the point $(0, 0, 0)$ has parametric equations

$$x = \lambda, \quad y = -\frac{3}{2}\lambda, \quad z = -2\lambda, \quad \lambda \in \mathbb{R}.$$

2 The inverse mapping theorem

The Dini's theorem for implicit functions of one variable can be used as follows in order to study the local invertibility of a given function of one variable. Let O be an open subset of $\mathbb{R}$ and $f : O \rightarrow \mathbb{R}$ be a function of class $\mathcal{C}^1$ such that

$$f(a) = b \quad \text{and} \quad f'(a) \neq 0. \quad (14)$$

Then there exists an open interval J containing b where the inverse function f^{-1} of f is defined. Moreover, f^{-1} has a continuous derivative defined by

$$f^{-1}(y) = \frac{1}{f'(f^{-1}(y))} \quad \forall y \in J. \quad (15)$$

In fact, the function $F(x, y) := f(x) - y$ satisfies $F(a, b) = 0$ and $F_x(a, b) \neq 0$. Therefore by Dini's theorem, there exists an open interval I centered at a and an open interval J centered at b and a unique function $g : I \rightarrow J$ of class $\mathcal{C}^1$ such that

$$f(x) = y \quad \Rightarrow \quad x = g(y) \quad \forall (x, y) \in I \times J$$

and

$$g'(y) = -\frac{F_y(g(y), y)}{F_x(g(y), y)} = -\frac{-1}{f'(g(y))} = \frac{1}{f'(g(y))} \quad \forall y \in J.$$

By definition the function g is denoted by f^{-1} .

In this section, we want to prove an analogous result for functions $f = (f_1, f_2, \dots, f_n) : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ as a consequence of Dini's theorem for systems.

Definition 2.1 Let $\Omega \subset \mathbb{R}^n$ be an open set and $f = (f_1, f_2, \dots, f_n) : \Omega \rightarrow \mathbb{R}^n$ be a function of class $\mathcal{C}^1$. We say that f is locally invertible around $a \in \Omega$ if there exists an open ball $B(a, \delta) \subset \Omega$ such that the restriction of f to $B(a, \delta)$ is invertible from $B(a, \delta) \rightarrow f(B(a, \delta))$. If such a restriction of class $\mathcal{C}^1$ together with its inverse, then we say that f is a local diffeomorphism around a .

Theorem 2.2 (Local invertibility)

Let $\Omega \subset \mathbb{R}^n$ be an open set, $f = (f_1, f_2, \dots, f_n) : \Omega \rightarrow \mathbb{R}^n$ be a function of class $\mathcal{C}^1$ and $a \in \Omega$. Assume that the jacobian of f is non zero at a :

$$\det \left(\frac{\partial(f_1, \dots, f_n)}{\partial(x_1, x_2, \dots, x_n)}(a) \right) \neq 0. \quad (16)$$

Then there exists an open neighborhood U of a and an open neighborhood V of $f(a)$ such that $f : U \rightarrow V$ is invertible and its inverse function $f^{-1} : V \rightarrow U$ is of class $\mathcal{C}^1$. Moreover, for any $y \in V$, we have

$$\frac{\partial((f^{-1})_1, (f^{-1})_2, \dots, (f^{-1})_n)(y)}{\partial(y_1, \dots, y_n)} = \left[\frac{\partial(f_1, f_2, \dots, f_n)}{\partial(x_1, \dots, x_n)}(f^{-1}(y)) \right]^{-1}. \quad (17)$$

Proof: Let $\delta_1 > 0$ be such that

$$B(a, \delta_1) \subset \Omega \quad \text{and} \quad \det \left(\frac{\partial(f_1, \dots, f_n)}{\partial(x_1, x_2, \dots, x_n)}(x) \right) \neq 0 \quad \forall x \in B(a, \delta_1). \quad (18)$$

Let us first prove that f is injective on $B(a, \delta_1)$. Assume by contradiction that there are $u, v \in B(a, \delta_1)$ such that $u \neq v$ while $f(u) = f(v)$. Then, since $\det \left(\frac{\partial(f_1, \dots, f_n)}{\partial(x_1, x_2, \dots, x_n)}(x) \right) \neq 0 \quad \forall x \in B(a, \delta_1)$, we get that

$$\left[\frac{\partial(f_1, \dots, f_n)}{\partial(x_1, x_2, \dots, x_n)}(x) \right] (u - v) \neq 0_{\mathbb{R}^n} \quad \forall x \in B(a, \delta_1) \quad (*)$$

Since the vector in the left-hand side of $(*)$ has components $\nabla f_i(x) \cdot (u - v)$ for $i = 1, \dots, n$, we get that

$$\nabla f_i(x) \cdot (u - v) \neq 0 \quad \text{for some } i \in \{1, \dots, n\}.$$

If we apply The Mean Value Theorem to the function $\phi : [0, 1] \rightarrow \mathbb{R}^n$ defined by $\phi(t) := f_i(u + t(v - u))$ we get a number $c \in (0, 1)$ such that

$$\phi(1) - \phi(0) = \phi'(c) \quad \Leftrightarrow \quad 0 = f_i(v) - f_i(u) = \nabla f_i(u + c(v - u)) \cdot (v - u)$$

which is impossible, since $u + c(v - u) \in B(a, \delta_1)$. Thus f is injective on $B(a, \delta_1)$.

Now let us prove that there exists $\delta_2 > 0$ such that $f(B(a, \delta_2))$ is an open set and f^{-1} is of class $\mathcal{C}^1$ on $f(B(a, \delta_2))$.

We consider the function

$$F(x, y) = f(x) - y. \quad (19)$$

Notice that $F(a, b) = f(a) - b = 0$ and

$$\det \left(\frac{\partial(F_1, \dots, F_n)}{\partial(x_1, x_2, \dots, x_n)}(a, b) \right) = \det \left(\frac{\partial(f_1, \dots, f_n)}{\partial(x_1, x_2, \dots, x_n)}(a) \right) \neq 0. \quad (20)$$

Hence, from Theorem 1.1, there exist $\delta, \beta > 0$ and a unique function $g : B(b, \beta) \rightarrow B(a, \delta)$ of class $\mathcal{C}^1$ such that

$$F(x, y) = 0 \Leftrightarrow x = g(y) \quad \forall y \in B(b, \beta). \quad (21)$$

That is,

$$y = f(x) \Leftrightarrow x = g(y) \quad \forall y \in B(b, \beta).$$

Now set

$$\delta_2 = \min(\delta_1, \delta) \quad U = B(a, \delta_2) \quad \text{and} \quad V = g^{-1}(B(a, \delta_2)).$$

So, since f is injective on $B(a, \delta_1)$, it follows that g is the inverse of the restriction of f on $U = g(V) = f^{-1}(V)$.

Now let us prove (17). It follows that

$$\frac{\partial(g_1, \dots, g_n)}{\partial(y_1, y_2, \dots, y_n)}(y) = - \left[\frac{\partial(f_1, \dots, f_n)}{\partial(x_1, x_2, \dots, x_n)}(g(y)) \right]^{-1} \left[\frac{\partial(F_1, \dots, F_n)}{\partial(y_1, y_2, \dots, y_n)}(g(y), y) \right]. \quad (22)$$

Now since

$$\frac{\partial F_i}{\partial y_j} = \begin{cases} 0 & \text{if } i \neq j, \\ 1 & \text{if } i = j, \end{cases},$$

it follows from (22) that

$$\frac{\partial((f^{-1})_1, \dots, (f^{-1})_n)}{\partial(y_1, y_2, \dots, y_n)}(y) = \left[\frac{\partial(f_1, \dots, f_n)}{\partial(x_1, x_2, \dots, x_n)}(f^{-1}(y)) \right]^{-1}. \quad (23)$$

■

Example 2.3 Consider the function $f = (f_1, f_2) : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$\begin{cases} f_1(x, y) := \ln(1 + x^2) + \cos(x - y) + y \\ f_2(x, y) := 2e^{x+2y} + x - y \end{cases}$$

- (a) Show that the function f is invertible in a neighborhood of the point $(0, 0)$.
- (b) Find the Jacobian matrix of the mapping f^{-1} at the point $(1, 2)$.

Solutions:

- (a) The function f is of class $\mathcal{C}^\infty$. Let us find the Jacobian matrix of the function f at the point $(0, 0)$.

$$\frac{\partial(f_1, f_2)}{\partial(x, y)}(x, y) = \begin{pmatrix} \frac{\partial f_1}{\partial x}(x, y) & \frac{\partial f_1}{\partial y}(x, y) \\ \frac{\partial f_2}{\partial x}(x, y) & \frac{\partial f_2}{\partial y}(x, y) \end{pmatrix}$$

$$\frac{\partial f_1}{\partial x} = \frac{2x}{1+x^2} - \sin(x-y), \quad \frac{\partial f_1}{\partial y} = \sin(x-y) + 1, \quad \frac{\partial f_2}{\partial x} = 2e^{x+2y} + 1, \quad \frac{\partial f_2}{\partial y} = 4e^{x+2y} - 1.$$

Hence,

$$\begin{aligned} \frac{\partial(f_1, f_2)}{\partial(x, y)}(0, 0) &= \begin{pmatrix} \frac{\partial f_1}{\partial x}(0, 0) & \frac{\partial f_1}{\partial y}(0, 0) \\ \frac{\partial f_2}{\partial x}(0, 0) & \frac{\partial f_2}{\partial y}(0, 0) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 3 & 3 \end{pmatrix} \\ \Rightarrow \det \left(\frac{\partial(f_1, f_2)}{\partial(x, y)}(0, 0) \right) &= \begin{vmatrix} 0 & 1 \\ 3 & 3 \end{vmatrix} = -3 \neq 0. \end{aligned}$$

Hence, by the inverse mapping theorem, there exists a neighborhood U of $(0, 0)$ and a neighborhood V of $f(0, 0) = (1, 2)$ such that the function $f : U \rightarrow V$ is invertible with an inverse $f^{-1} = ((f^{-1})_1, (f^{-1})_2) : V \rightarrow U$ of class $\mathcal{C}^\infty$.

We say that the function f is a **diffeomorphism of class $\mathcal{C}^\infty$** around the point $(0, 0)$.

- (b) From the inverse mapping theorem, if $f(x, y) = (s, t)$, then the Jacobian matrix of the inverse mapping $f^{-1} = ((f^{-1})_1, (f^{-1})_2)$ at the point $(1, 2) = f(0, 0)$ is equal to

$$\frac{\partial((f^{-1})_1, (f^{-1})_2)}{\partial(s, t)}(1, 2) = \left(\frac{\partial(f_1, f_2)}{\partial(x, y)}(0, 0) \right)^{-1} = \begin{pmatrix} 0 & 1 \\ 3 & 3 \end{pmatrix}^{-1} = -\frac{1}{3} \begin{pmatrix} 3 & -1 \\ -3 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 1/3 \\ 1 & 0 \end{pmatrix}$$

The next set of notes will be on invertible linear mappings.